

Graph-Based Molecular Generation via Adversarial Learning with Hierarchical Discriminators and Reward-Driven Objectives

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Abstract—Recent advances in deep-generative models have allowed the generation of chemically valid and novel molecules with desirable properties for further investigation. In this work, we provide such a model, integrating generative adversarial networks (GANs), hierarchical graph discrimination, and reinforcement learning. Our graph-based generator generates molecular graphs in the form of adjacency and node feature matrices. To improve robustness and stability during training, we implement a hierarchical differentiable pooling (DIFFPOOL)-based discriminator. This type of discriminator allows our model to learn coarse-grained graph representations during training. Additionally, we introduce a reinforcement learning component in which a policy network receives reward signals based on predicted chemical properties, guiding the generator toward structures with high drug-likeness and other target metrics. In evaluating our approach, we look for validity, uniqueness, novelty, and alignment with drug-likeness metric scores. The results demonstrate improved generation quality and training stability compared to baseline graph GANs, particularly when incorporating DIFFPOOL and reinforcement-guided reward shaping. Our findings suggest that the combination of hierarchical discrimination with reinforcement learning shows a positive trajectory for property-driven molecular design in drug discovery and materials science.

Index Terms—GAN, Hierarchical discriminator, DIFFPOOL, Reinforcement learning, Validity, Uniqueness, Novelty, Drug-likeness

Nomenclature

$A \in \mathbb{R}^{n \times n}$	adjacency matrix
$T = [t_1, t_2, \dots, t_n] \in \mathbb{Z}^n$	atomic numbers (node features)
$B \in \mathbb{Z}^{n \times n}$	bond order matrix (optional)
$G = (V, E)$	molecular graph with vertices V and edges E

I. INTRODUCTION

In our research, we used the QM9 dataset to train a GAN generator working alongside a hierarchical discriminator to generate chemically valid and novel molecules for scientific study. We used two different versions of the hierarchical dis-

criminator, one with DIFFPOOL, and one without. In association with the generator and discriminator, we also implement a reward policy network to encourage the model’s generation. The dataset we used, QM9, contained the following variables: mol_id, smiles, A, B, C, mu, alpha, homo, lumo, gap, r2, zpve, u0, u298, h298, g298, cv, u0_atom, u298_atom, h298_atom, g298_atom, Morgan_fingerprint, adj_matrix, node_features, and node_count. The variables that are significant for this work are adj_matrix, node_features, and node_count. These variables define the molecules that are fed into the generator during training. The adjacency matrix, adj_matrix, contains information on which bonds are present inside the molecule. The node features, node_features, contain information on which atoms are present within the molecule. In this study, we analyze the validity, novelty, uniqueness, and drug-likeness of molecules produced by the generator. The validity is based on whether or not the generated molecule is chemically valid, uniqueness is based on similarity of the generated molecule to previous molecules, novelty is determined by whether or not a similar molecule is already present within the training data, and drug-likeness is concerned with the similarity of the molecule generated to common drug-like molecules.

II. LITERARY REVIEW

III. METHODS

A. Adjacency Matrix to Molecule

1) *Vertex Set (Atoms)*: Each atom is represented by a vertex:

$$V = \{v_i = \text{Atom}(t_i) | i = 1, 2, \dots, n\}$$

Optionally, if chirality tags are given:

$$\text{Chiral}(v_i) = c_i \text{ for } i \in C \subseteq \{1, \dots, n\}$$

2) *Edge Set (Bonds)*: For all $i < j$:

$$\text{If } A_{ij} > \tau \text{ (threshold, e.g., 0.5) :}$$

$$o_{ij} = \begin{cases} B_{ij}, & \text{if } B_{ij} > 0 \\ 1, & \text{otherwise} \end{cases}$$

$$E \ni e_{ij} = (v_i, v_j, \beta(o_{ij}))$$

Where $\beta(o)$ maps bond order to bond type:

$$\beta(o) = \begin{cases} \text{SINGLE}, & o = 1 \\ \text{DOUBLE}, & o = 2 \\ \text{TRIPLE}, & o = 3 \\ \text{AROMATIC}, & o = 4 \\ \text{undefined}, & \text{otherwise} \end{cases}$$

3) *Isolated Hydrogen Handling*: Let:

$$H = i | t_i = 1 \wedge \deg(v_i) = 0$$

For each $h \in H$:

$$j^* =$$

IV. EASE OF USE

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Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

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Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \quad (1)$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use “(??)”, not “Eq. (??)” or “equation (??)”, except at the beginning of a sentence: “Equation (??) is . . .”

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Please use “soft” (e.g., `\eqref{Eq}`) cross references instead of “hard” references (e.g., (1)). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

Please don’t use the `{eqnarray}` equation environment. Use `{align}` or `{IEEEeqnarray}` instead. The `{eqnarray}` environment leaves unsightly spaces around relation symbols.

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- There is no period after the “et” in the Latin abbreviation “et al.”.
- The abbreviation “i.e.” means “that is”, and the abbreviation “e.g.” means “for example”.

An excellent style manual for science writers is [?].

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TABLE I: Table Type Styles

Table Head	Table Column Head		
	Table column subhead	Subhead	Subhead
copy	More table copy ^a		

^aSample of a Table footnote.



Fig. 1: Example of a figure caption.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity “Magnetization”, or “Magnetization, M”, not just “M”. If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write “Magnetization (A/m)” or “Magnetization {A[m(1)]}”, not just “A/m”. Do not label axes with a ratio of quantities and units. For example, write “Temperature (K)”, not “Temperature/K”.

ACKNOWLEDGMENT

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